

A systematic study of Nasicon-type $\text{Li}_1 + x\text{M}_x\text{Ti}_2 - x(\text{PO}_4)_3$ (M: Cr, Al, Fe) by neutron diffraction and impedance spectroscopy



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ABSTRACT

A systematic study of $\text{Li}_1 + x\text{M}_x\text{Ti}_2 - x(\text{PO}_4)_3$ with M = Al, Cr and Fe and $0 \leq x \leq 1$ has been carried out by X-ray and neutron powder diffraction and by impedance spectroscopy. The analysis of the diffraction data allowed us to describe the disposition of the lithium atoms within the structure. They were found to occupy two different positions, the well-known M1 and a new position called M2'. The addition of trivalent metals into the $\text{LiTi}_2(\text{PO}_4)_3$ compound, even in small amount, gives rise to the liberation of the M1 sites and simultaneous occupation of the M2' sites. This situation becomes more pronounced as the substituent content increases, meaning a higher delocalization of lithium atoms within the structure. Moreover, rising temperature also provokes the movement of lithium from M1 to M2'. The impedance spectroscopy measurements allowed us to relate the electrical properties to the observations of the crystal structure. Both dc bulk and grain boundary ionic conductivity show a thermally activated behavior, and their values increase with substituent content until compositions of $x = 0.2$ – 0.3 . The activation energy for bulk dc conductivity hardly changes with the substitution, and the main factor affecting the variation of the ionic conductivity is the pre-exponential factor. We find indeed a correlation between the pre-exponential factor and the degree of disorder of the lithium atoms in the structure as determined from neutron diffraction data. The highest conductivity was found in the sample $\text{Li}_{1.3}\text{Al}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ with a value of $6.2 \times 10^{-3} \text{ S cm}^{-1}$ at room temperature and activation energy of 0.30 eV, which is comparable with those measured in the best ionic conductors reported to date.

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1. Introduction

Lithium-based solid electrolytes have been extensively studied in the last decades for their application in batteries and fuel cells [1–7]. Specifically, its high potential and low weight makes lithium the best candidate for high-energy density storage batteries. In the search for fast lithium ionic conductors, different materials have been optimized. Nasicon-type materials (Na Superionic CONductor) were first discovered as three-dimensional sodium ionic conductors [8]. Since then, many studies have focused on the development of Nasicon materials with optimized ionic conductivity and high stability. In particular, NASICON-type materials with Li^+ ions occupying the open channels in the structure were investigated as promising candidates for lithium batteries. $\text{LiTi}_2(\text{PO}_4)_3$ and its derivatives are among the best options due to the small size of the Ti^{4+} that makes the size of the sites in the channels more appropriate for lithium ions [9–15].

The crystal structure of $\text{LiTi}_2(\text{PO}_4)_3$ is based on the Nasicon rhombohedral structure (S.G. $R\bar{3}c$) and can be described as formed by infinite ribbons of $[\text{Ti}_2(\text{PO}_4)_3]^-$ units with [102] orientation. These ribbons are composed of alternate corner-sharing $[\text{TiO}_6]$ octahedra and $[\text{PO}_4]$ tetrahedra. As reported in the literature, Li^+ ions may occupy two interstitial positions within this network, the so-called M1 site, **6b** (0, 0, 0), and the M2 sites, **18e** (x , 0, 1/4). The M1 site is sixfold coordinated by oxygen atoms, and the M2 site has an eightfold coordination. Both sites alternate along the channels forming a three-dimensional conduction network. The different connectivity makes the relative occupancy of M1 and M2 sites a key factor to determine the ionic transport of the NASICON materials. When all the M1 sites are occupied by lithium, all the pathways for long-range ion motion are blocked by the own ionic carriers, as in the case of $\text{LiTi}_2(\text{PO}_4)_3$ compound, where Li^+ ions occupy preferentially the low connectivity M1 positions [16,17]. The appropriate substitution of Ti^{4+} with trivalent metals like Al^{3+} , Fe^{3+} or Cr^{3+} has successfully improved the ionic conductivity in this system as a consequence of different factors. It promotes the occupancy of M2 sites, liberating part of the formerly blocked M1 sites and expanding the ionic conduction pathways in such a way that ions can now percolate the sample [14]. Also, it produces an increase in the carrier (Li^+) content, and it enhances

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the lithium conductivity at grain boundaries due to higher densification during material processing [18,19].

Several studies describe the disposition of lithium atoms in $\text{Li}_{1+x}\text{M}_x\text{Ti}_2 - x(\text{PO}_4)_3$ compounds studied by neutron powder diffraction (NPD) and nuclear magnetic resonance (NMR), but only selected stoichiometries have been reported, and to our best knowledge, no systematic study exists to date [14,20–23]. In this work, powders of composition $\text{Li}_{1+x}\text{M}_x\text{Ti}_2 - x(\text{PO}_4)_3$ for $0 \leq x \leq 1$ have been synthesized using a sol–gel synthesis route. A systematic study of the structural characteristics and electrical properties of the prepared samples has been carried out using X-ray powder diffraction (XRPD), neutron powder diffraction (NPD) and impedance spectroscopy. Herein, we report the structural changes that take place in these systems as a consequence of the migration of lithium ions and their influence on the electrical properties.

2. Experimental

2.1. Sample preparation

$\text{Li}_{1+x}\text{M}_x\text{Ti}_2 - x(\text{PO}_4)_3$ samples with $0 \leq x \leq 1$ and $\text{M} = \text{Al}, \text{Cr}$ and Fe were prepared by the Pechini method [24]. LiNO_3 , $\text{Fe}/\text{Cr}/\text{Al}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, $\text{NH}_4\text{H}_2\text{PO}_4$ and $\text{TiC}_8\text{H}_{24}\text{O}_4$ were used as starting materials with citric acid (CA) as complexing and ethyleneglycol (EG) as polymer agents. The molar ratios of CA:M and EG:CA were set to 2 and 4, respectively. $\text{TiC}_8\text{H}_{24}\text{O}_4$ was initially added to a saturated solution of CA whose pH was previously adjusted to 8, to promote ionization and complexation of titanium. After total dissolution, the rest of the reactants were added to the solution, and subsequently, the appropriate amount of EG was mixed with the solution. The temperature was then increased to 413 K to promote polyesterification. During heating, solutions became more viscous until a polymeric gel was formed. The obtained gel was heated up to 673 K until a dark powder precursor was obtained. The resultant precursors were heated at 1073 K during 4 h. This yielded samples of composition $\text{Li}_{1+x}\text{M}_x\text{Ti}_2 - x(\text{PO}_4)_3$ with selected values of x in the range of $0 \leq x \leq 1$ (termed LTP for Al, LCTP for Cr and LFTP for Fe samples followed by the x value in the general formula).

2.2. Structural characterization

XRPD patterns of the synthesized powders were collected at room temperature in a PANalytical X'Pert PRO instrument using $\text{CuK}\alpha_1$ radiation with a PW 3050/60 goniometer. Continuous scan with a virtual step size of 0.033° (2θ) and a counting time of 350 s was employed in the range of $2\theta = 10$ – 120° . The instrumental parameters were previously calibrated using the LaB_6 standard NIST SRM 660b.

High-resolution neutron diffraction experiments were carried out at selected temperatures between 77 and 540 K using the instruments HRPT [25] and E9 [26] of the European facilities PSI and HZB, respectively. Approximately 3 g of the sample was placed in a vanadium cylindrical holder with a thickness of 6 mm. At HRPT, data were collected in a 2θ range of 4 to 165° with a step of 0.05° and $\lambda = 1.494 \text{ \AA}$. At E9, the range 3° to 142° with integration steps of 0.07° was measured with $\lambda = 1.799 \text{ \AA}$. X-ray and neutron diffraction data were analyzed using the Rietveld method implemented in the Fullprof software [27]. The neutron powder diffraction data were refined, according to the Nasicon-type structure, starting with a structural model with the atoms that make up the skeleton of the structure, i.e., Ti/M, P, O1 and O2, in 12c (0, 0, 0.14), 18e (0.29, 0, 1/4), 36f (0.18, 0.99, 0.19) and (0.19, 0.16, 0.19) sites, respectively. Atomic coordinates were then refined while the profile parameters were kept constant. Based on this model, different Fourier's maps were calculated. These maps show the scattering density difference between the real structure and the model and, from them, the positions of the atoms missing in the starting model—lithium atoms—can be determined. Since lithium has negative coherent scattering length for neutrons, it shows up as negative scattering density in Fourier's maps.

2.3. Ionic conductivity measurements

The powder samples were pressed into pellets of 13 mm diameter and 1 mm of thickness using a uniaxial pressure of 150 MPa and subsequently sintered at 1273 K. The density of the pellets was estimated from the measurements of their mass and volume. The electrical conductivity of the sintered pellets was analyzed by impedance spectroscopy. Measurements were carried out in a BDS80 from Novocontrol, in the frequency range 10^{-2} – 10^7 Hz at selected temperatures between 123 and 523 K with the integrated Quatro temperature control system with accuracy of ± 0.1 K. Electrical contacts were made with silver paint and an inert atmosphere was ensured by measuring the samples under N_2 flow.

3. Results and discussion

3.1. X-ray diffraction at room temperature

The phase composition of samples $\text{Li}_{1+x}\text{M}_x\text{Ti}_2 - x(\text{PO}_4)_3$ with $0 \leq x \leq 1$ and $\text{M} = \text{Al}, \text{Cr}, \text{Fe}$ was studied by XRPD. Fig. 1 shows the XRPD patterns of samples $\text{Li}_{1+x}\text{M}_x\text{Ti}_2 - x(\text{PO}_4)_3$ with $0 \leq x \leq 1$. Single phase materials with rhombohedral Nasicon structure (SG: $R\bar{3}c$) were obtained when $x \leq 0.4$ in the aluminum system and when $x \leq 0.3$ in chromium and iron systems. At higher x values, some additional reflections were found marked in Fig. 1. In the LTP system, small maxima found in the samples with $x = 0.5$ and 0.7 were assigned to AlPO_4 with SG: Pc [PDF:00-051-1674]. This phase has been observed to segregate in similar compounds by other authors [12]. When $x = 1$, the sample crystallizes in a different crystal structure of lower symmetry that was ascribed as monoclinic Nasicon with SG: $P2_1/n$ [PDF: 0111--078-1106] that was previously reported in $\text{LiZr}_2(\text{PO}_4)_3$ [28]. When $\text{M} = \text{Cr}$ and Fe , the monoclinic Nasicon phase is formed as secondary phase when $x = 0.4, 0.5$, with increased intensity, and finally as single phase when $x = 1$.

3.2. Neutron diffraction as a function of temperature

Samples observed as single phases in Fig. 1, i.e., with $x \leq 0.4$ when $\text{M} = \text{Al}$ and $x \leq 0.3$ when $\text{M} = \text{Cr}$ and Fe , were studied by powder neutron diffraction at room temperatures. Three selected samples, LTP, LTP03 and LCTP02, were also studied at variable temperatures from 77 to 540 K. The first stage of the Rietveld data treatment was the refinement of the structural skeleton, i.e., Ti/M, P and O atoms as described in the experimental part. Different Fourier maps were then calculated. Fig. 2 shows, as an example, the different Fourier maps in the xy plane with $z = 0$ obtained for the LTP samples at selected temperatures. In these maps, the M1 position described in the literature (0,0,0) is seen as a well-defined negative scattering density in the LTP sample at 77 K. The increase of the temperature gives rise to the expansion of the negative scattering density. On the other hand, the substitution of Ti^{4+} by trivalent metals also produces the expansion of the negative scattering density in the M1 position that is completely delocalized when $x \geq 0.3$ (Fig. 2). Lithium atoms were then added to the model in the M1 position. The high values of thermal coefficients (B) obtained and the delocalization observed in the Fourier maps led us to refine them anisotropically. The anisotropic B coefficients of the Li^+ atoms in M1 brought out a preference thermal displacement on the xy plane and negligible along the z direction due to the presence of the $[\text{Ti}/\text{MO}_6]$ octahedra. Other authors described this finding as the movement of atoms from M1 to a general position around [20]. Equivalent behavior was found with temperature and substitution in Cr and Fe systems.

Further analysis of the different Fourier maps revealed negative scattering density due to the additional Li^+ atoms, with the maximum located at position (0.47(3), 0.31(1), 0.25(1)) in all samples except in LTP (see Fig. S1). This position is close to the M2 (0.53, 0, 1/4) described in the literature but displaced in b direction from the centre of the polyhedron. In the following, this position will be termed M2'. Li^+

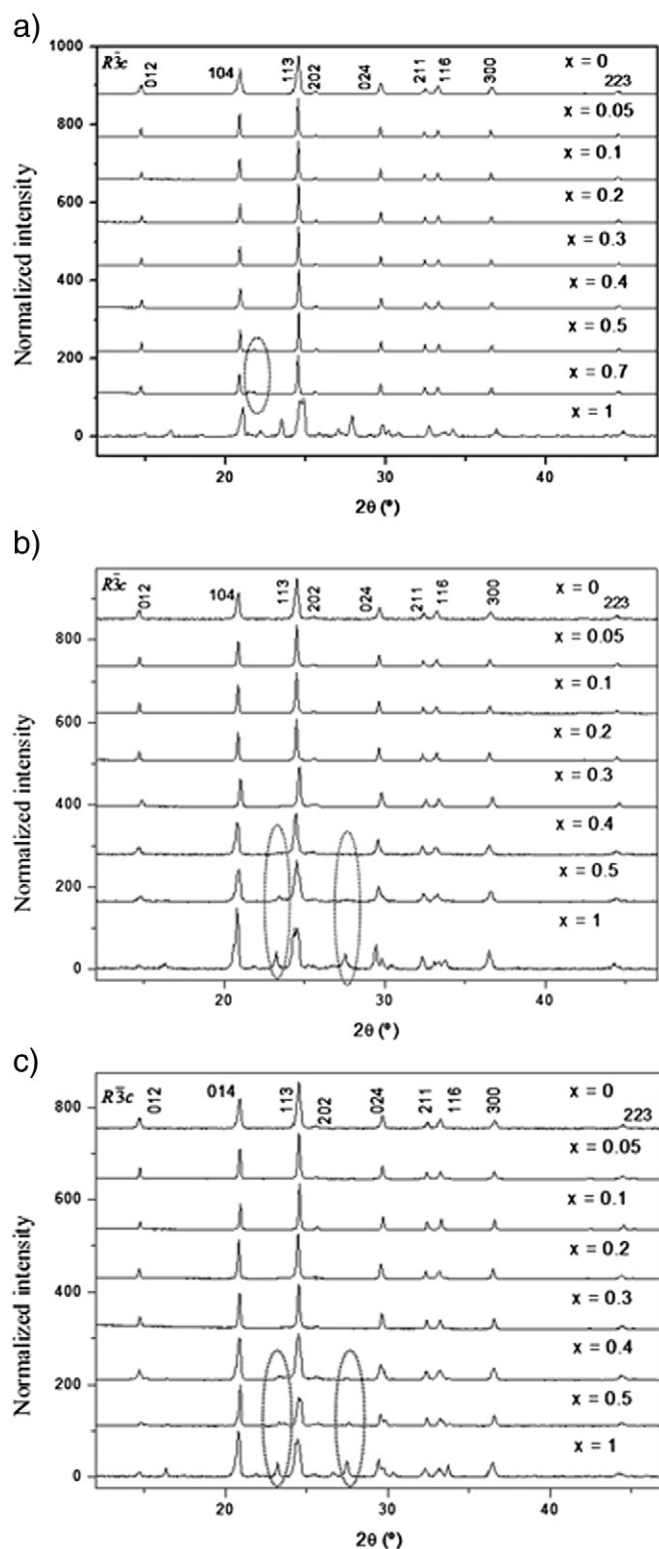


Fig. 1. XRPD patterns obtained at room temperature in samples $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ with $0 \leq x \leq 1$ and (a) $\text{M} = \text{Al}$, (b) $\text{M} = \text{Cr}$ and (c) $\text{M} = \text{Fe}$. Dashed lines indicate the presence of secondary phases (AlPO_4 in LATP and Nasicon monoclinic in LCTP and LFTP systems).

atoms in $\text{M}2'$ position ($\text{Li}2'$) were added to the structural model during the data analysis. The atomic coordinates and site occupation factors (s.o.f.) of $\text{M}1$ and $\text{M}2'$ were refined constraining the B coefficients to the isothermal ones obtained for $\text{Li}1$. The s.o.f. of $\text{Li}1$ and $\text{Li}2'$, as well as M and Ti , were also constrained to have the sum according to the expected stoichiometry. Fig. 3 shows, as an example, the fitted neutron diffraction pattern of LATP01 sample at room temperature.

In Fig. 4, the unit cell is represented along the conduction path $\text{M}1-\text{M}2'-\text{M}1$ and the Li^+ atoms in $\text{M}1$ ($\text{Li}1$) and $\text{M}2'$ ($\text{Li}2'$) are indicated. In Fig. 4, the displacement of the $\text{M}2'$ position from the previously reported data is noticeable. The fact that we observed a displaced position of $\text{M}2$ together with the highly anisotropic thermal displacement of Li^+ in $\text{M}1$ is an evidence of the high mobility of lithium atoms within the structure.

Table 1 shows the refined cell parameters, atomic coordinates and occupation factors of Li atoms, as well as the agreement factors of each sample at room temperature and two of them at different temperatures. Additional information (atomic coordinates, Biso and s.o.f. of the atoms which shape the structural skeleton) is shown in Tables S1 and S2 of the supplementary materials. It is worth noting that the obtained occupation factors of M/Ti show that the degree of substitution is in all cases somewhat lower than expected from stoichiometry, with increasing tendency toward higher substituent content. This is an indication of a small amount of a secondary phase. In fact, on the sample LFTP03, a small amount of the monoclinic phase of $\text{Li}_2\text{FeTi}(\text{PO}_4)_3$ (SG: $P2_1/n$) was found (3.6(2) % wt.), which was not detected previously in the XRPD pattern. The experimental values of their cell parameters, $a \approx 8.602(1)$ Å, $b \approx 11.986(2)$ Å, $c \approx 8.533(1)$ Å and $\beta \approx 90.052(5)$, are in good agreement with the values reported in the literature [29].

In general, the variation in the cell parameters is in good agreement with the ionic radii of the different substituents ($r(\text{Al}^{3+}) = 0.535$ Å, $r(\text{Ti}^{4+}) = 0.605$ Å, $r(\text{Cr}^{3+}) = 0.615$ Å, $r(\text{Fe}^{3+}) = 0.645$ Å) [30], increasing with the x value in Cr and Fe systems and decreasing in the Al system. Although there are other factors affecting the cell parameters, such as the disposition of Li^+ ions, it has been reported that the ionic radii of substituents is the fundamental limitation [31,32], which proves that in fact substitution of Ti^{4+} takes place. From the values of the site occupation factors in Table 1, it can be deduced that the Li^+ ions are displaced from $\text{M}1$ site to $\text{M}2'$ site with increasing the x value, which results in a higher delocalization of lithium within the structure, since the $\text{M}2'$ sites are of bigger size. In fact, in the case of Cr and Fe systems, when $x \geq 0.2$, the fraction of ions in $\text{M}2'$ is higher than in $\text{M}1$, giving rise to an intermediate situation between $\text{LiTi}_2(\text{PO}_4)_3$ and $\text{Li}_3\text{M}_2(\text{PO}_4)_3$, but with a higher degree of disorder [33,34]. Especially significant is the liberation of ions from $\text{M}1$ that is observed in the samples with lowest x values, with site occupation factors varying from 0.167 (when $x = 0$) to 0.114, 0.125 and 0.083 (when $x = 0.05$) and $\text{M} = \text{Al}$, Cr and Fe , respectively (Table 1).

The diffraction experiments made at different temperatures let us to study the thermal expansion of the samples. The thermal expansion coefficients of materials with similar composition have been reported to be dependent on the synthesis method [35,36]. Fig. 5 shows the variation of cell parameters of different samples at various temperatures. As can be seen, the variation of c lattice parameter with temperature is significant, while a remains nearly constant (varies less than 0.2%). Thermal expansion coefficients were calculated as $\alpha = \frac{l_2 - l_1}{l_1(T_2 - T_1)}$, where l_1 and l_2

are the cell parameters at temperatures T_1 and T_2 , respectively, and values of $\alpha_c \sim 1.5 \times 10^{-5} \text{ K}^{-1}$ and $\alpha_a \sim 0$ were obtained. From the analysis of the bond distances, the size of the $\text{M}1$ cavity has been found to increase along the c direction and decrease along a , with temperature. This behavior, previously observed by Alamo et al. [37], provokes cooperative rotations of $[\text{TiO}_6]$ octahedra and $[\text{PO}_4]$ tetrahedra. On the other hand, the obtained values of the occupation factor of Li^+ in $\text{M}1$ and $\text{M}2'$ shows a small displacement from $\text{M}1$ to $\text{M}2'$ with temperature.

3.3. Impedance spectroscopy

The charge dynamics of the samples was investigated by complex impedance measurements. The wide ranges of frequency and temperature used have allowed us to separate the bulk dc and grain boundary mechanisms of the electrical conductivity. Fig. 6a shows, as an example, the plot of the real part of the conductivity versus frequency of the

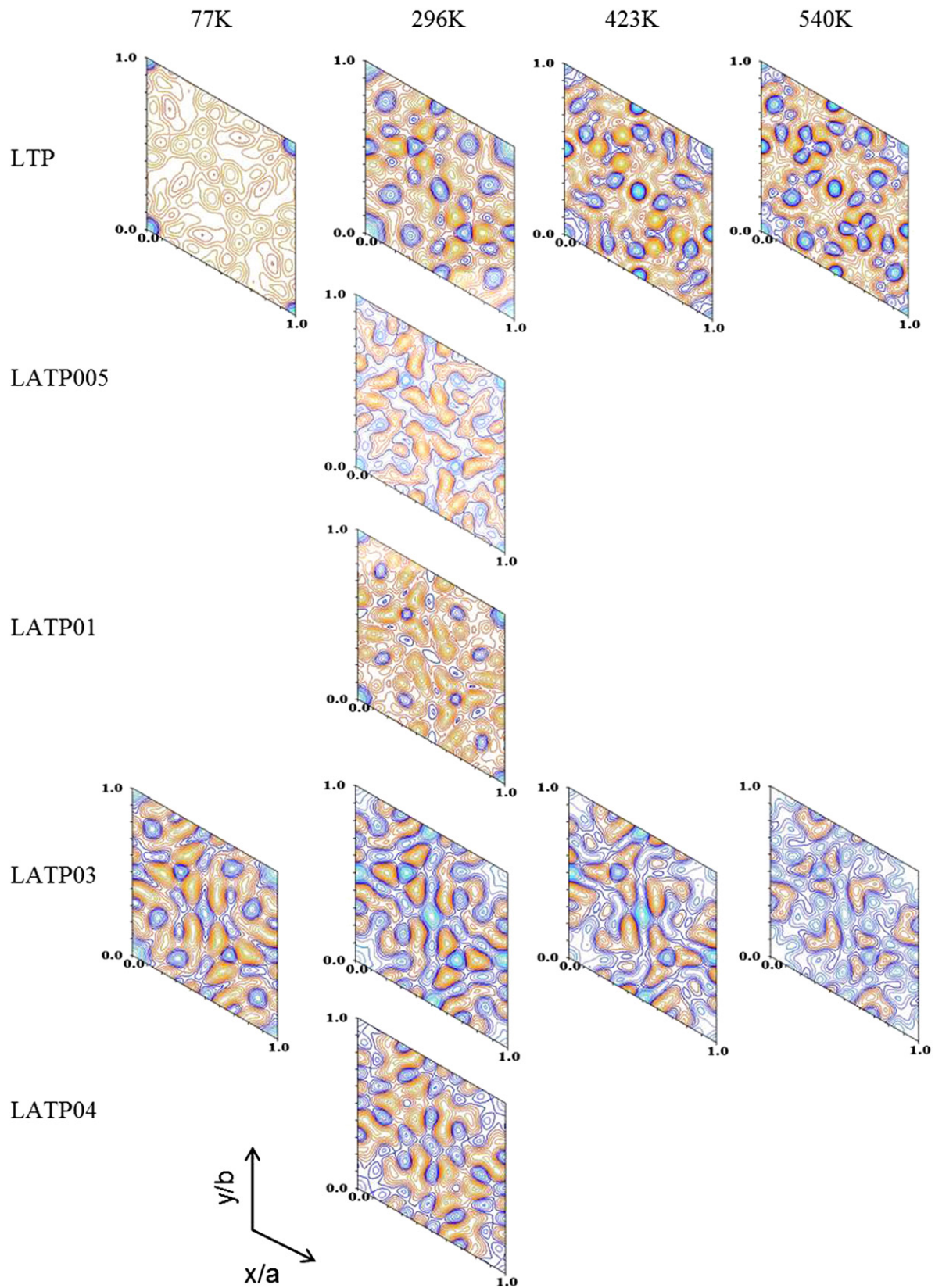


Fig. 2. Different Fourier maps in the xy plane with $z = 0$ of the neutron diffraction data at selected temperatures, obtained from Rietveld refinement of the Ti/Al-P-O skeleton of the samples $\text{Li}_{1+x}\text{Al}_x\text{Ti}_{2-x}(\text{PO}_4)_3$. Remaining scattering length densities are shown blue when negative, orange when positive.

representative LFTP03 sample. The spectra of all the samples show the typical features of ionic transport: (i) a dispersive regime in the low-frequency range at the highest temperatures due to the blocking

of the mobile ions in electrodes, (ii) a plateau at higher frequencies corresponding to grain boundary electrical transport followed by a dispersive regime and (iii) a new plateau due to the long-range or dc bulk

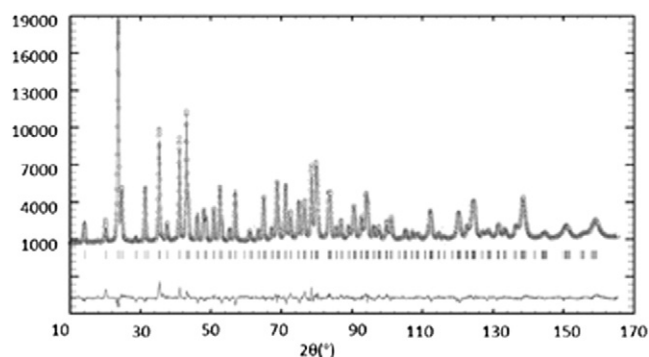


Fig. 3. Experimental, calculated and different NPD patterns obtained at room temperature of the sample LAPT01; bars indicate the position of Bragg peaks.

ionic conductivity, which is followed again by a dispersive regime at the highest frequencies. In Fig. 6b, the Nyquist plot, Z'' versus Z' , of the same sample at 163 K is shown where the three different conductive regimes described above can be also distinguished: (i) a straight line at high Z' (low-frequency values) shows the blocking of ionic carriers in the metallic electrodes, (ii) a big semicircle at medium frequencies evidences the grain boundary conductivity and (iii) a small semicircle close to the origin (high frequency, see inset of Fig. 6b) shows the bulk contribution to the conductivity. In some samples, especially in the aluminum ones, the different conductivity regimes—bulk and grain boundary—are difficult to distinguish using these representations. The overlap of both semicircles in these kinds of samples arises because bulk and

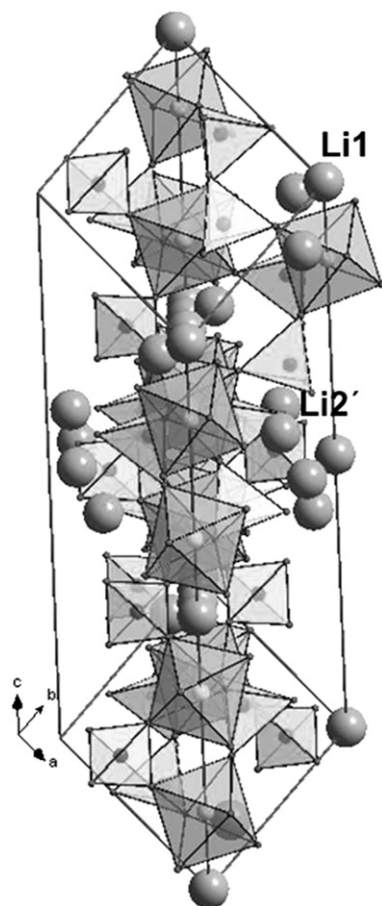


Fig. 4. Crystal structure of samples $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ along the conduction path, M1-M2'-M1 , where the M1 and M2' sites are shown.

grain boundary contributions to the conductivity are very close in frequency. This can be ascribed to low electrical homogeneity of the samples and makes it necessary to use different representations of the conductivity data [38,39]. The electrical modulus is simply related to the permittivity, $M^* = 1/\epsilon^*$, and it is a very convenient representation in this case because the spectrum is dominated by the lowest capacity element and therefore the maximum at each temperature describes the bulk response of the sample [40,41]. Fig. 7 shows the frequency dependence of the imaginary part of the electrical modulus for LFTPO3 sample at different temperatures. The maxima of the spectra are asymmetric, evidencing the dispersion of the electrical conductivity at the high frequency side of the peak and indicating the existence of correlation effects in the conductivity [42–45]. These correlation effects are a consequence of the existence of interactions among the moving ionic carriers (ion–ion interactions) since they are charged and have to compete for the vacant positions in the structure, giving rise a non-Debye relaxation process [46]. The frequency value at the peak corresponds to the end of the plateau of the dc conductivity in the conductivity representation and can be used as a criterion to determine the bulk dc conductivity. Using the combination of both representations, we are able to extract the bulk and grain boundary conductivity values at different temperatures for each sample and represent it in an Arrhenius plot. We calculated the long-range or dc conductivity σ_{dc} , as the σ' value at the precise frequency of the peak of the modulus. The grain boundary values σ_{gb} are much easier to determine since this regimen extends several decades in frequency in all the samples, as it is shown in Fig. 6a, and can be extracted from the plateau in the conductivity representation. The results are shown in Fig. 8 for selected samples. Lines in Fig. 8 are fits to the equation $\sigma_{dc} \cdot T = \sigma_0 \exp(-E_a/k_B T)$, where σ_0 is the pre-exponential factor, E_a is the activation energy and k_B is the Boltzman constant. All samples follow the Arrhenius law, and from the fitting, we obtained E_a , σ_0 and conductivity values at room temperature of each mechanism, see Table 2.

As it is noticeable in Table 2, the electrical conductivity generally increases with substitution. Especially important is the increase observed with the lowest levels of substituents, $x = 0.05$, compared with the sample with $x = 0$. The highest conductivity values were found for $x = 0.3$ in aluminum and iron substituted samples and for $x = 0.2$ in the chromium samples, corresponding to the lowest activation energy of grain boundaries, which approach the values of the bulk. In systems where the inter- and intra-grain activation energy is similar, differences in conductivity are normally related to the connectivity between grains, which have a higher concentration of imperfections near the grain boundary [47]. In this work, the maximum conductivity is found in the sample $\text{Li}_{1.3}\text{Al}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ ($\sigma_b = 6.2 \times 10^{-3} \text{ S cm}^{-1}$ and $\sigma_{gb} = 7.3 \times 10^{-5} \text{ S cm}^{-1}$ at room temperature), which is in fact the most studied composition of the $\text{LiTi}_2(\text{PO}_4)_3$ derivatives. However, other authors found maxima values in samples with $x = 0.2$ or $x = 0.5$ [14,26].

As can be seen in Table 2, grain boundary conductivity values are two or three orders of magnitude smaller than those in the bulk. The small conductivity of grain boundary is related to the low density of the samples. The calculated density varies from 45% (LTP) to 70% (LATPO3) of the theoretical density. The highest values were found in LTP samples. In the three systems studied, the density is increased by increasing the substitution level.

The variation of activation energy with composition in the bulk is smaller than in the grain boundaries. This small variation may be due to changes in the size of the electrochemical window in the conduction path. Li^+ ions placed in M1 site are surrounded by six oxygen atoms inside an antiprism and the jump to M2' sites implies passing through the lateral face of this site which constitutes the bottleneck of the conduction path [48]. Table 2 shows that the nature of the substituent does not affect significantly the size of the bottleneck because the activation energy of bulk conduction decreases with the substitution content in the three systems independently of the ionic radii of substituent. Thus,

Table 1

Refined results of cell parameters, atomic coordinates and site occupation factor of Li⁺ ions and agreement factors for different samples at room temperature. Data obtained at different temperature for samples LATP03 and LCTP02 is also added in *italics*.

	a (Å)	c (Å)	Li1 s.o.f.	Li2' s.o.f.	x	y	z	R _b	R _{wp}
LTP	8.5087(1)	20.9089(4)	0.167	–	–	–	–	2.34	8.01
LATP005	8.5066(1)	20.9085(3)	0.114(4)	0.061(4)	0.202(9)	0.033(10)	0.089(3)	3.00	8.70
LATP01	8.5068(1)	20.8905(3)	0.116(3)	0.067(3)	0.199(8)	0.034(8)	0.089(3)	3.18	8.50
LATP03	8.4941(1)	20.8586(4)	0.113(5)	0.104(5)	0.143(12)	0.036(10)	0.104(6)	2.87	8.57
77 K	<i>8.4886(1)</i>	<i>20.7839(5)</i>	<i>0.115(5)</i>	<i>0.102(5)</i>	<i>0.213(11)</i>	<i>0.010(8)</i>	<i>0.071(3)</i>	2.70	<i>9.08</i>
423 K	8.4918(1)	20.9125(1)	0.105(6)	0.111(6)	0.215(11)	0.015(9)	0.070(4)	3.95	12.9
540 K	8.4939(1)	20.9750(5)	0.081(6)	0.136(6)	0.196(10)	0.006(8)	0.066(3)	4.27	12.2
LATP04	8.4882(2)	20.8571(7)	0.065(8)	0.168(8)	0.203(9)	0.018(8)	0.075(3)	4.05	11.7
LCTP005	8.5093(1)	20.9150(5)	0.125(5)	0.050(5)	0.232(14)	0.033(19)	0.072(7)	2.36	8.58
LCTP01	8.5126(1)	20.9212(4)	0.112(4)	0.071(4)	0.197(8)	0.034(8)	0.091(3)	3.14	8.77
LCTP02	8.5132(2)	20.9271(6)	0.063(6)	0.137(6)	0.198(9)	0.025(8)	0.073(3)	2.86	9.82
77 K	<i>8.5043(2)</i>	<i>20.8405(7)</i>	<i>0.062(7)</i>	<i>0.138(7)</i>	<i>0.201(6)</i>	<i>0.026(8)</i>	<i>0.077(3)</i>	2.82	<i>9.85</i>
423 K	8.5064(2)	20.9712(7)	0.066(7)	0.134(7)	0.209(9)	0.029(10)	0.075(4)	3.86	12.2
540 K	8.5090(2)	21.0332(9)	0.047(10)	0.153(10)	0.214(10)	0.029(11)	0.081(4)	4.28	14.5
LCTP03	8.5183(3)	20.9177(10)	0.071(6)	0.140(5)	0.198(9)	0.016(9)	0.071(4)	3.50	13.3
LFTP005	8.5125(1)	20.8871(5)	0.083(5)	0.092(5)	0.225(8)	0.028(10)	0.073(4)	2.60	8.49
LFTP01	8.5137(1)	20.8822(3)	0.107(3)	0.076(3)	0.198(7)	0.034(7)	0.090(2)	3.02	8.45
LFTP02	8.5213(1)	20.9211(5)	0.082(5)	0.118(5)	0.217(8)	0.010(9)	0.075(3)	2.47	9.64
LFTP03	8.5239(1)	20.9215(4)	0.099(3)	0.117(5)	0.187(6)	0.019(7)	0.090(2)	2.83	8.86

the variation of the activation energy is due to the different location of Li⁺ ions, changing from M1 to M2' sites as demonstrated by NPD. When M1 sites are empty, repulsion forces between oxygen atoms are increased, extending the M1 site and increasing the size of the bottleneck [49].

Considering the different substituents present in the samples, the highest bulk conductivity values have been obtained in the aluminum samples, where the lowest activation energy and highest pre-exponential factors were found. Looking at the activation energy, the lowest values have been found in the aluminum system followed by chromium and iron. This sequence agrees with the increase of the ionic radii, which seems to be the determinant factor. On the other hand, the variation of the pre-exponential factor follows the sequence Al > Fe > Cr. The pre-exponential factor can be described as:

$$\sigma_0 = \frac{4\alpha e^2 \alpha^2 v_0 N e^S / k_B}{k_B}$$

where α is a geometrical factor, a is the jump distance, v_0 is the attempt frequency, N is the concentration of carriers, S is a configurational entropy term, e is the electron charge and k_B is the Boltzmann constant. Since the

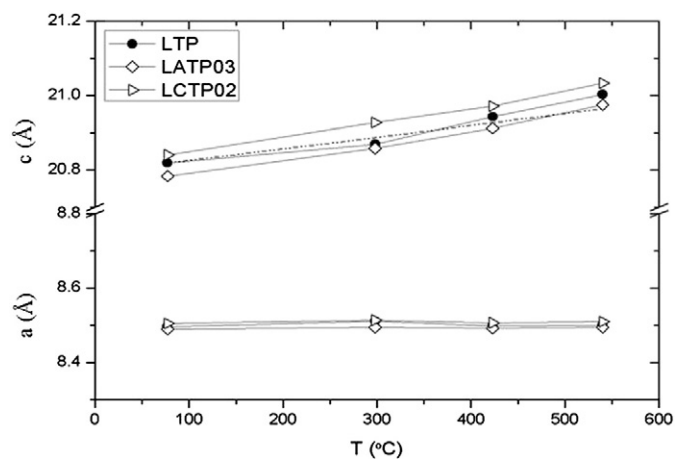


Fig. 5. Variation of cell parameters with temperature. Theoretical variation with $\alpha = 1.5 \times 10^{-5} \text{ K}^{-1}$ is added as dot line.

structure remains extremely similar and we are very slightly varying lithium content, we consider the configurational entropy term as the most probable cause of the differences observed in Table 2, over one order of magnitude [50]. Taking into account the occupation factors obtained from the neutron diffraction data, Li⁺ ions are displaced to the M2' sites with substitution. The relation of the occupation factors $|M1-M2'| / (M1 + M2')$ is thus a measure of the order in the lithium

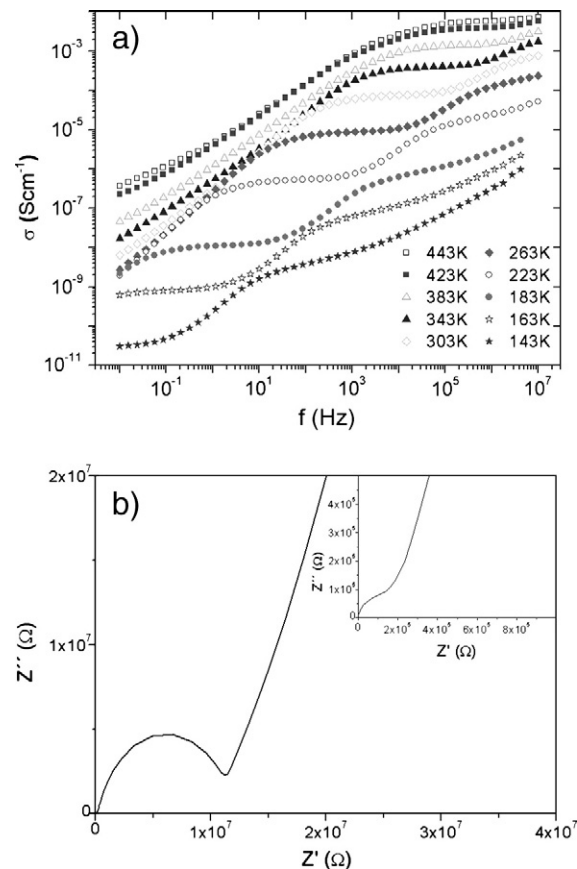


Fig. 6. (a) Conductivity versus frequency and (b) Nyquist plot at 163 K of the sample $\text{Li}_{1.3}\text{Fe}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$.

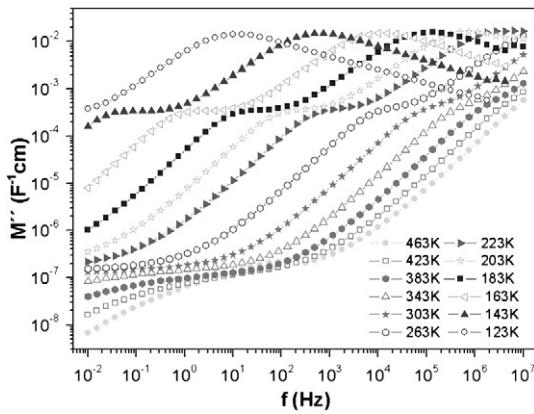


Fig. 7. Modulus versus frequency plot of sample $\text{Li}_{1.3}\text{Fe}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$.

occupation in the structure. Thus, the configurational entropy increases with the disorder for Li^+ ions and takes its maximum when the value of the order parameter $|M1-M2'| / (M1 + M2')$ is zero, i.e., $M1$ and $M2'$ are equal. According to this relation, the lowest ratio $|M1-M2'| / (M1 + M2')$ is for the sample LATP03, which has the higher pre-exponential factor of the series. Fig. 9 shows that a correlation indeed exists between the values of the pre-exponential factor and the degree of order in the structure for all the samples, independent of the substitution level. There is a clear trend showing the exponential increase of the pre-exponential factor with increasing disorder.

4. Conclusions

Nasicon-type samples of composition $\text{Li}_{1+x}\text{M}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ with $x \leq 0.4$ ($M = \text{Al}$), $x \leq 0.3$ ($M = \text{Cr}$) and $x \leq 0.2$ ($M = \text{Fe}$) were successfully synthesized as pure phases at 1073 K using the Pechini synthesis method. The substitution of Ti by trivalent metals in of $\text{LiTi}(\text{PO}_4)_3$ forces the Li^+ ions to occupy a new position different from $M1$, which have been determined by NPD as $M2'$. Even at low substitution levels ($x = 0.05$), the release of $M1$ and simultaneous occupation of $M2'$ has been observed. On the other hand, the increase in temperature also provokes the movement of Li^+ ions from $M1$ to $M2'$. Impedance spectroscopy measurements show a slight variation of the activation energy for dc bulk conductivity that is attributed to the different location of Li^+ ions, changing from $M1$ to $M2'$ sites. The main factor affecting the ionic conductivity is the pre-exponential factor of the Arrhenius formula. The observed trend of

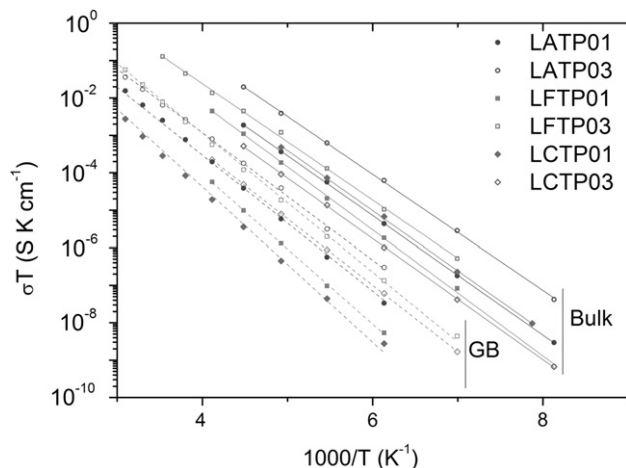


Fig. 8. Arrhenius fitting of selected samples in the bulk and grain boundary regions.

Table 2

Activation energy, pre-exponential factor and ionic conductivity at room temperature in the bulk and grain boundaries obtained from Arrhenius fit.

Sample	BULK			G.B.		
	E_a (eV) ± 0.01	σ_0 (S cm^{-1}) $\pm 30\%$	σ_{RT} (S cm^{-1}) $\pm 30\%$	E_a (eV) ± 0.01	σ_0 (S cm^{-1}) $\pm 30\%$	σ_{RT} (S cm^{-1}) $\pm 30\%$
LTP	0.33	5.8×10^3	6.0×10^{-5}	0.51	3.3×10^4	3.0×10^{-7}
LATP005	0.31	5.6×10^4	1.1×10^{-3}	0.26	4.2	6.5×10^{-7}
LATP01	0.32	4.7×10^4	6.7×10^{-4}	0.38	2.4×10^4	2.8×10^{-5}
LATP02	0.31	1.9×10^5	4.0×10^{-3}	0.34	5.9×10^3	3.6×10^{-5}
LATP03	0.30	1.7×10^5	6.2×10^{-3}	0.33	7.5×10^3	7.3×10^{-5}
LATP04	0.29	7.3×10^4	3.3×10^{-3}	0.38	4.1×10^4	5.2×10^{-5}
LCTP005	0.33	1.8×10^4	1.8×10^{-4}	0.47	1.1×10^4	4.6×10^{-7}
LCTP01	0.32	1.7×10^4	2.6×10^{-4}	0.41	8.3×10^3	3.1×10^{-6}
LCTP02	0.31	2.7×10^4	4.3×10^{-4}	0.36	8.6×10^3	2.1×10^{-5}
LCTP03	0.32	1.8×10^4	2.9×10^{-4}	0.35	4.4×10^3	1.6×10^{-5}
LFTP005	0.34	1.7×10^4	1.3×10^{-4}	0.47	2.4×10^4	1.0×10^{-6}
LFTP01	0.34	9.5×10^4	6.3×10^{-4}	0.40	9.3×10^3	6.0×10^{-6}
LFTP02	0.32	1.0×10^5	1.4×10^{-3}	0.29	6.9×10^2	2.9×10^{-5}
LFTP03	0.31	1.1×10^5	2.3×10^{-3}	0.37	3.0×10^4	6.0×10^{-5}
LFTP05	0.32	1.9×10^4	2.7×10^{-4}	0.41	8.8×10^3	3.3×10^{-6}

the pre-exponential factors is due to differences in the configurational entropy term due to different degree of order of the Li^+ ions in the structure, that is expressed as the ratio of the site occupation factors $|M1-M2'| / (M1 + M2')$.

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.ssi.2014.07.018>.

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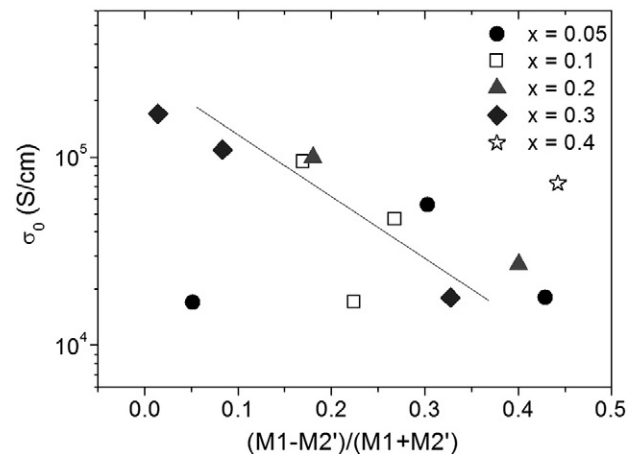


Fig. 9. Pre-exponential factors determined from the Arrhenius fits of the bulk dc conductivity for the different samples as a function of the order parameter $|M1-M2'| / (M1 + M2')$ in site occupation factors as determined from neutron diffraction data. Different symbols correspond to different substitution level x in the samples.

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